

Clustering Cities by Weather

By: Sreeja Pagireddy and Svetlana Nosova

Background

- We are investigating the similarity between cities based on wind speed and air temperature.
- In this analysis, we are using dendrogram to show how cities can be grouped based on weather similarity.
- This analysis may help provide people with information on which cities have similar climate and climate-related issues, such as energy usage for heating or cooling.
- It can also help people compare cities when considering relocation.

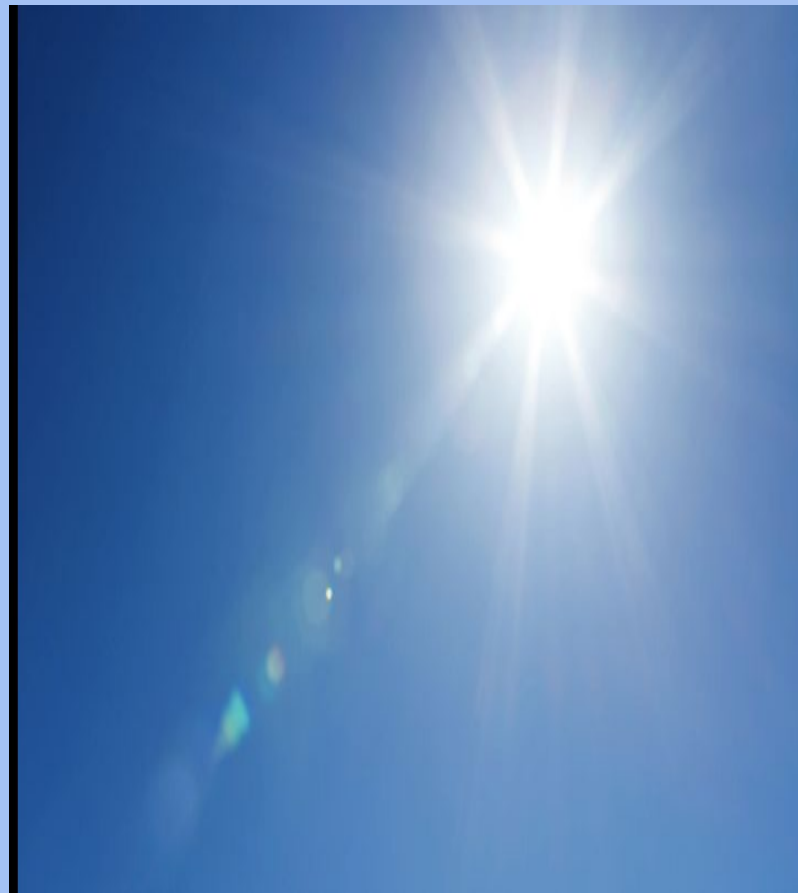


Image source:

<https://www.thecalifornian.com/story/news/my-planet/2016/05/10/salinas-valley-weather-sunny/84177760/>

Discovery Expectations

- We don't expect to see a uniform distribution of the data points and there may be several outliers that influence the dataset.
- We believe there might be several clusters that are very different from each other.
- We expect to find several groups of cities with similar temperature and wind speed.



Image source:

<https://dailycampus.com/2023/03/22/roundtable-cold-weather-vs-warm-weather-which-is-better/>

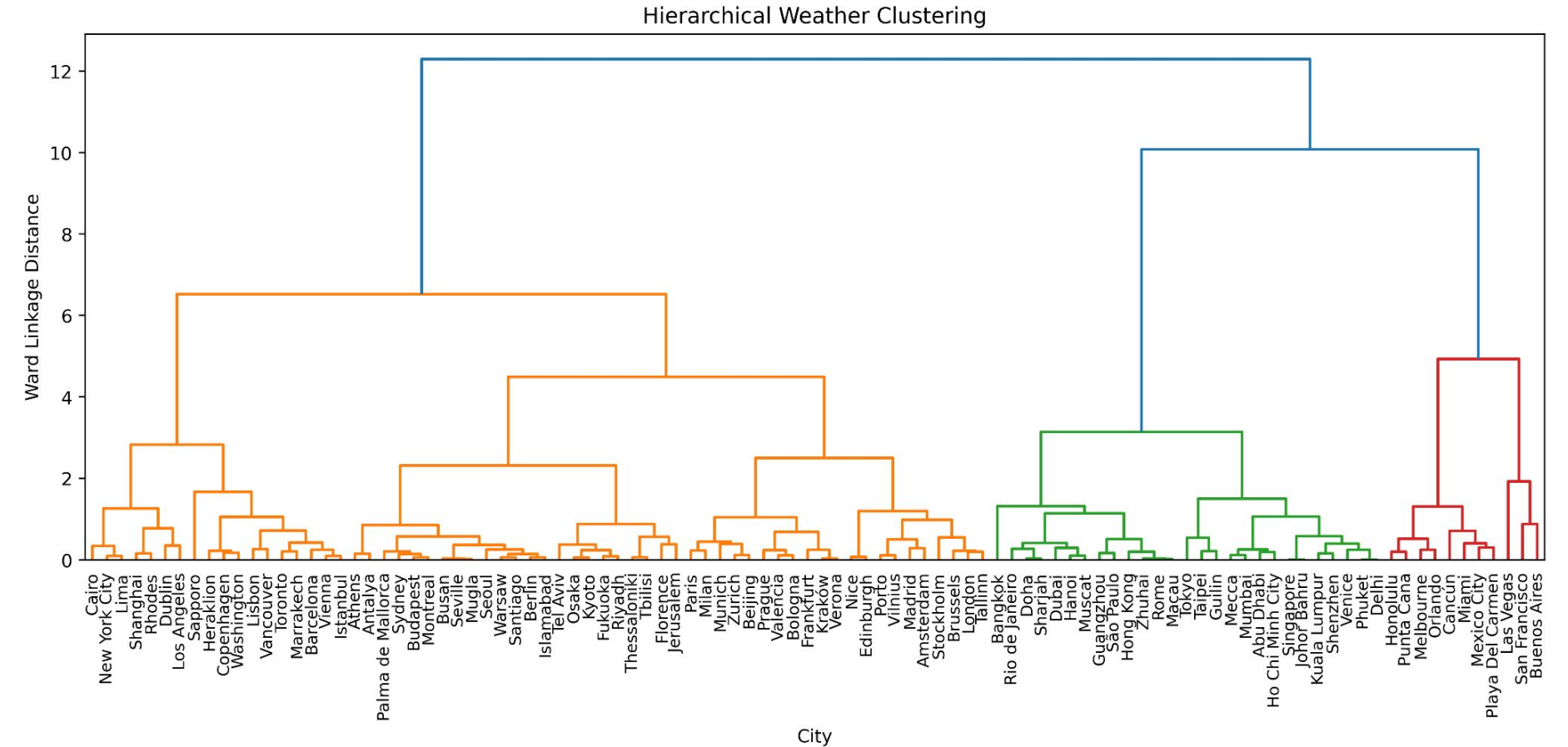
Feature Usage

- Dataset we used: <https://www.kaggle.com/danishammar/top-100-cities-weather-dataset>
- The dataset contains several columns. Since a dendrogram requires numerical input only, we kept numerical columns and did not include categorical features in the analysis. Latitude and longitude were also excluded, as they are the geographical features, not weather features.
- Numerical features used for clusterization are: the average temperature and the average wind speed recorded on April 28, 2024.
- Z score normalization was used on the features to scale the data.
- Before running a hierarchical clustering, we checked the data for missing and duplicate values.

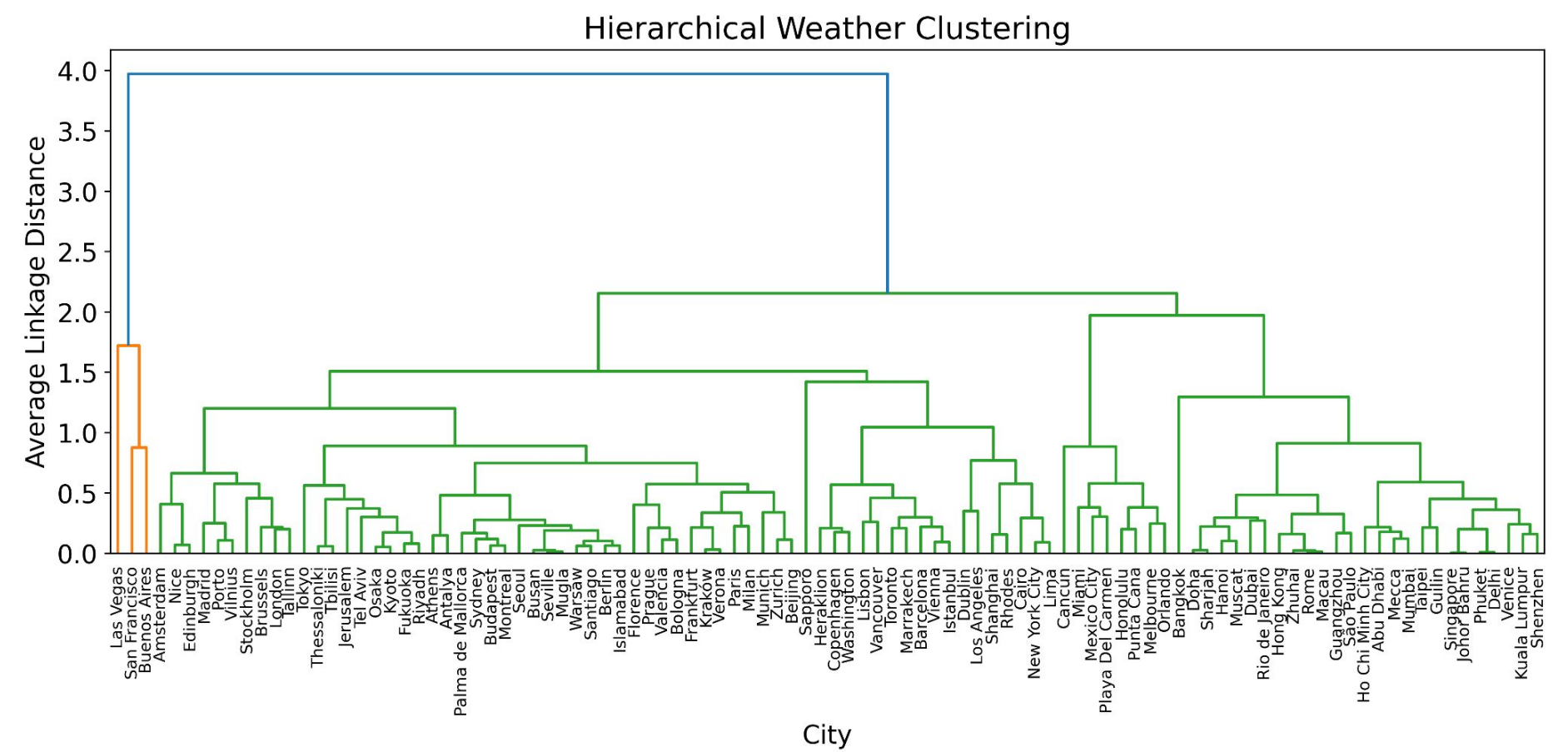
Distance Measure and Linkage function

- Euclidean distance was used to measure similarity between cities.
- The Ward and Average methods was used as a linkage function.
- Ward produced a visualization by making the clusters more compact, as this method keeps cities in the same cluster as similar as possible without increasing the variance inside a cluster.
- Average helped us make sure the results in Ward aren't completely random.

Results



Results



Observations

- Kyoto, Osaka, and Tokyo are grouped closely together, which supports the geographic and climatic reasons of the clustering.
- Zurich, Munich, Milan, and Paris are also close to each other, which is supported by their geographical closeness.
- London and Edinburgh remain in the same subgroup in both dendrograms, showing a stable pattern across the two linkage methods.
- The Ward green cluster represents cities from subtropical and desert regions, while the orange cluster includes cities from Europe and North America.
- The Average method produces less clear and compact clusters and groups.

Limitations

- The dendrogram is difficult to read when there are too many cities.
- We used only two features, so the clusters do not fully describe the climate in each city.
- Only one-day snapshot was analyzed.
- Different distance and linkage methods may produce different results.

Future Work

- Include more features such as humidity, air pressure, or AQU for a deeper analysis.
- Analyze more data.
- Compare more distances and linkage methods to optimize our analysis and produce better results.
- Expand the analysis by comparing weather similarity not only between the cities but also between countries or regions.

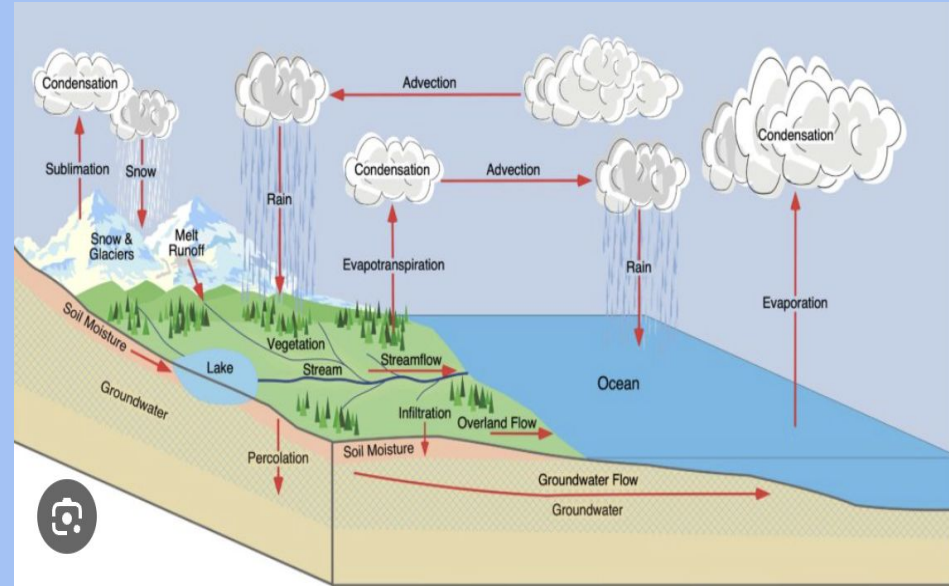


Image source:

<https://pressbooks.bccampus.ca/physgeoglabmanual1/chapter/lab8/>

Sources

<https://medium.com/@whyamit101/pandas-only-keep-certain-columns-7710d6f71a56>

<https://python-graph-gallery.com/400-basic-dendrogram/>

<https://gist.github.com/vals/150ec97a5b7db9c82ee9>

Thank You

Any Questions?